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McGranahan et al.

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(54) **WALNUT ROOTSTOCK ‘VX211’**

(50) Latin Name: *Juglans hindsii*×*Juglans regia*
Varietal Denomination: **VX211**

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A01H 5/00 (2006.01)

(52) **U.S. Cl.** **Plt./154**

(58) **Field of Classification Search** **Plt./154**
See application file for complete search history.

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(57) **ABSTRACT**

A new and distinct variety of walnut rootstock denominated ‘VX211’ is described. This new variety, ‘VX211’, can be propagated through standard tissue culture micropropagation or rooted cuttings. ‘VX211’ has vigor and survivability in the nursery and in the orchard. It has reduced susceptibility to damage from nematodes (*Pratylenchus vulnus*) compared to other ‘Paradox’ rootstock. ‘VX211’ also has reduced susceptibility to damage from *Phytophthora citricola* in greenhouse screens and in the field compared to other ‘Paradox’ rootstock.

10 Drawing Sheets

BACKGROUND OF INVENTION

The present invention relates to a new and distinct clonal rootstock of 'Paradox' (not patented) walnut tree (*Juglans hindsii* × *Juglans regia*) that has been denominated vari-
 5 as 'VX211', and more particularly to such a walnut rootstock that is vigorous, that has reduced susceptibility to nematodes (*Pratylenchus vulnus*) and *Phytophthora* (*Phytophthora citricola*), and that further is easily clonally propagated by stan-
 10 dard tissue culture micropropagation.

It has long been recognized as desirable to provide vigor-
 ous walnut rootstocks that have increased resistance to soil
 pests and diseases, specifically nematodes (*Pratylenchus vul-*
nus) and *Phytophthora* (*Phytophthora citricola*). The root-
 stock of the present variety, 'VX211', is similar to other
 15 'Paradox' walnut rootstocks (*Juglans hindsii* × *Juglans regia*), except that it has increased resistance or tolerance to *Praty-*
lenchus vulnus and *Phytophthora citricola* and can be easily
 micropropagated to produce a vigorous clonal 'Paradox' root-
 stock.

SUMMARY OF THE INVENTION

It was found that the walnut rootstock 'VX211' of the
 present invention exhibits the following combination of char-
 25 acteristics:

- a) can be propagated through standard tissue culture micro-
 propagation or rooted cuttings;
- b) has vigor and survivability in the nursery and in the
 orchard;
- c) has reduced susceptibility to damage from nematodes
 (*Pratylenchus vulnus*) compared to other 'Paradox' root-
 stock; and
- d) has reduced susceptibility to damage from *Phytoph-*
thora citricola in greenhouse screens and in the field
 compared to other 'Paradox' rootstock.

BRIEF DESCRIPTION OF THE TABLES

Table 1 shows comparative nursery performance of
 40 'VX211' and other walnut rootstock clones.

Table 2 shows growth rating and diameters for 'VX211'
 and other rootstock clones.

Table 3 shows field performance of clonal 'Paradox' wal-
 nut hybrids, Northern California black walnut and Chinese
 wingnut rootstocks in non-infested soil and soil infested with
Phytophthora.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows performance of 'VX211' compared to 'AX1'
 under pressure from nematodes at varying densities.

FIG. 2 shows a visual rating of tree growth and condition of
 clonal and seedling test trees at a Jenny Lind, Calif. field site.

FIG. 3 shows cumulative percent mortality of clonal and
 seedling test trees at a Jenny Lind, Calif. field site.

FIG. 4 shows tree mortality of clonal and seedling test trees
 at a Linden, Calif. field site.

FIG. 5 shows relative resistance to *Phytophthora citricola*
 among clonal and seedling test trees in greenhouse experi-
 60 ment.

FIG. 6 shows 'VX211' at 10 years of age.

FIG. 7 shows bark of ten-year old 'VX211'.

FIG. 8 shows several three-year old trees of 'VX211'.

FIG. 9 shows bark at base of three-year old 'VX211'.

FIG. 10 shows bark and lenticels farther up the stem of
 'VX211'.

FIG. 11 shows 6-month old greenhouse grown 'VX211'.

FIG. 12 shows grafted 'VX211'.

FIG. 13 shows spring foliage of 'VX211'.

FIG. 14 shows deep red of new foliage of 'VX211'.

FIG. 15 shows upper side of mature leaf of 'VX211'.

FIG. 16 shows lower side of mature leaf of 'VX211'.

DETAILED DESCRIPTION OF THE INVENTION

The new rootstock of the present invention was selected as
 part of the "Paradox Diversity Study" (PDS) which was ini-
 tiated in 1996 to study the genetic diversity of commercial
 'Paradox' sources. The study included approximately
 300–500 seeds each (depending on the predicted percent
 'Paradox'), from 37 black walnut sources of 'Paradox' pro-
 15 vided by California walnut nurseries, and seven controlled
 crosses made in Davis, Calif. and open-pollinated controls
 from different *Juglans* species. Seeds were germinated and
 grown at 3 different nurseries for one year and then seedlings
 were distributed to cooperating researchers for tests of
 response to nematodes, *Phytophthora* (seed supplied), crown
 gall (*Agrobacterium tumefaciens*) and the field environment
 20 (field trials). The study was repeated in 1997.

The rootstock of the present invention was evaluated for
 response to nematodes in 1998 along with 9 siblings and the
 remaining rootstock families. One-year old seedlings were
 planted on 1.2 or 1.8 meter spacing with 3.35 meter centers.
 30 The field test site was infested with a single population of root
 lesion nematode (*P. vulnus*) originally placed on site in 1976.
 At planting time seedlings were inoculated with additional *P.*
vulnus. Each fall 20 grams of root tissue were collected from
 each tree. These roots were placed in a mist chamber for 5
 35 days for nematode extraction and nematodes/gram root was
 calculated.

In July 1998 it was evident that one seedling ('VX211')
 was more vigorous than the others, but in the fall the nema-
 todes were abundant on the roots of 'VX211'. In July 1999
 roots were collected and again nematodes were found to be
 abundant, but the seedling 'VX211' continued to be more
 vigorous than the other seedlings in spite of the nematodes.
 Due to its apparent superiority it was transplanted to a
 "mother block" at an agriculture center in Parlier, Calif. In
 winter 2000–2001 propagating wood was collected. 'VX211'
 45 was propagated by hardwood cuttings. Additionally,
 'VX211' was asexually reproduced by standard tissue culture
 micropropagation in Davis, Calif. In 2002 a "stock block"
 was established in Davis, Calif. with 6 trees of 'VX211' as
 well as other promising selections.

Propagation of 'VX211' both by standard hardwood cut-
 tings and by standard tissue culture micropropagation was
 successful. From November 2002 to October 2003, 212
 'VX211' plants were micropropagated, rooted in gelled
 55 medium and acclimatized in the greenhouse; 153 (72%) sur-
 vived. When rooted ex vitro, 126/184 (68%) survived. Hard-
 wood cuttings rooted between 73% (11/15) to 87% (13/15).
 By November 2003 there were 187 available for field trials
 and 60 available for *Phytophthora* screening.

In March 2004, 48 plants of the 'VX211' clone and a
 standard 'Paradox' ('AX1') were evaluated for additional
 nematode screening and comparison. These were planted in
 1/100th acre macroplots. The macroplots had concrete sides 1.5
 65 meter deep into the soil with open bottoms and were nema-
 tode-free. 'VX211' and 'AX1' were planted side by side in 48

separate macroplots infested with 0, 1, 20, or 500 *P. vulnus* nematodes per 250 cc of soil. Tree diameters and number of nematodes on the roots were determined 2004–2006 (FIG. 1). Nematodes built up quickly but ‘VX211’ was 30% taller in the first year and the diameter of ‘VX211’ was significantly greater than ‘AX1’ in all three years (FIG. 1). The vigor of ‘VX211’ under pressure from nematodes suggests that ‘VX211’ has a means to avoid or escape severe damage from nematodes.

In spring 2004, 106 plants of ‘VX211’ produced through standard tissue culture micropropagation were planted in a nursery along with over 1800 plants of 17 different clones. At the end of the growing season ‘VX211’ was the most vigorous of all clones (Table 1) demonstrating that propagation is true-to-type through successive generations. Eighty-two percent were graftable and the mean diameter was 31 mm at 5 cm from the soil surface.

Graftable trees were distributed for grafted field trials in 5 different orchards in replant situations in 2005. Grafting posed no problem and ‘VX211’ is considered compatible with English walnut scions. It is a typical ‘Paradox’ in that respect. In addition, 30 each of 11 different genotypes including ‘VX211’ were planted in May 2005 in Davis, Calif. for artificial inoculation with *Phytophthora citricola*. A randomized block split plot design was used. For each rootstock clone, there were six four-tree plots to be infested and six single-tree plots to serve as non-infested controls. Northern California black (*Juglans hindsii*) and wingnut (*Pterocarya stenoptera*) were included as susceptible and resistant controls, respectively. In January 2006, 100 ml of a V8 juice-oat mixture infested with *Phytophthora citricola* was mixed into the upper 5 cm of soil around the trunk of each tree. A sterile mixture was applied to the uninoculated controls.

Early results from several of the grafted field trials are shown (FIGS. 2–4; Table 2). In all cases ‘VX211’ was one of the superior clones. The block artificially inoculated with *Phytophthora* was assessed for growth in trunk circumference and development of crown rot as indicated by trunk cankers extending up from the soil surface in November 2006. Sixty-two percent of the susceptible controls were rotted or dead. The uninoculated controls of ‘VX211’ were the most vigorous trees in the block apart from the wingnut controls. No cankers were found on ‘VX211’ or many of the other clones, however the *Phytophthora* inoculation did appear to depress growth somewhat in ‘VX211’ (Table 3).

Simultaneously with field trials, greenhouse trials were carried out to assess the relative susceptibility of ‘VX211’ and other selected clones to *Phytophthora citricola*. Standard *Phytophthora* screening methods were used. Clonal selections including ‘VX211’ were micropropagated, rooted, acclimatized and chilled, and at 2–6 months were transplanted to pots of artificially inoculated soil. Four isolates of *P. citricola* from different districts of California were used to infest the soil. The isolates were grown in separate jars of V8 juice-oat-vermiculite substrate for one month, mixed in equal proportions and mixed in the soil (40 ml inoculated substrate per liter of soil). Starting two weeks after transplanting, all plants received 48-hour periods of soil flooding every two weeks. Three months after transplanting, soil was washed from the plants and the incidence and severity of crown rot

were determined. One selected clone, ‘VX211’ consistently showed moderate resistance. The results from 2006 trials are shown in FIG. 5.

BOTANICAL DESCRIPTION OF THE PLANT

This description is based on the original selection of ‘VX211’, ungrafted, at ten years of age, a 3-year old ungrafted tree in *Phytophthora* field screen, and a greenhouse-grown plant at 6 months of age. Figures are also shown of a grafted ‘VX211’ tree. Data for the botanical description were collected in spring 2007.

The Munsell Color Charts for Plant Tissues (1977. Gretag-Macbeth, New Windsor N.Y.) is used in the identification of color. Also, common color terms are to be accorded their ordinary dictionary significance,

Botanical classification: *Juglans hindsii* × *Juglans regia*.

Female parent: *Juglans hindsii*.

Male parent: *Juglans regia*.

‘VX211’ differs from its female parent by having fewer leaflets/leaf, broader leaflets and hybrid vigor. ‘VX211’ differs from its male parent by having more leaflets/leaf and hybrid vigor. ‘VX211’ does not differ substantially from other similar hybrids except in its superior performance under adverse soil conditions as described in the “Background of the Invention”.

Plant: The growth habit of the tree is illustrated in FIG. 6. This 10 year old tree is approximately 7.3 meters in height with a canopy diameter of approximately 5 meters. The trunk circumference at 1.2 meters above ground level is about 0.61 meters. The bark and year-old branches are light brown (2.5Y 7/2) (FIG. 7). New shoots are green (5GY 7/6). Lenticels (approximately 12 per 2.5 square cm) are slightly lighter than the bark (2.5Y 8/2). The 3 year old trees are 4.3–4.9 meters tall (FIG. 8). The bark is brownish-green (2.5GY 5/8) with scattered (22/2.5 square cm.) buff-colored lenticels (7.5YR 8/2) (FIGS. 9, 10). The six month old greenhouse grown tree is about 35 cm. tall and the main stem is about 1 cm. in diameter (FIG. 11) and green (5GY/10). Lenticels about 0.5 mm long are more dense at the base of the plant and are a buff color (2.5Y 8/4). Graft take is in the normal range for seedling ‘Paradox’ walnuts (FIG. 12).

Foliage: The slightly pubescent new spring foliage has a reddish hue to it (10R 4/8), darkest towards the tip (FIGS. 13 and 14) turning green (5GY 5/6) as the leaves get older. The leaves are smooth and the margins are entire (not serrate). The leaves are pinnately compound with 13–15 leaflets. The mature leaves of the 6 month old plant have 9–11 leaflets and are 30 cm long and 23 cm wide. The number of leaflets may vary depending on the age and size of the plant. The upper leaf surface is bright green and the same color as the stem (5GY 5/10) (FIG. 14). The lower surface is slightly duller (5GY 6/6) (FIG. 16). The leaflets are about 5 cm wide and 14 cm long with a petiole 4–8 cm long.

Inflorescence: No catkins or female flowers appeared in the first 10 years. The tree is probably male sterile as is typical with *Juglans hindsii* × *Juglans regia* hybrids. No nuts were observed.

Disease resistance and susceptibility: This rootstock is typical of other *Juglans hindsii* × *Juglans regia* hybrids except that it possesses higher vigor and ability to survive heavy nematode loads. It is also less susceptible to *Phytophthora citricola* than other similar hybrids.

Usage: The new rootstock of the present invention provides walnut growers with a new clonally propagated ‘Paradox’ rootstock. It can be easily micropropagated.

TABLE 1

Clones grown in Stanislaus County, California in 2004							
Clone	Planted	Graft-able	Graft-able	Diameter (mm)			
	N	N	%	Mean	SD	Range	CV
<u>Nematodes</u>							
VX211	106	87	82	31	4.9	21-44	12.6
<u>Phytophthora</u>							
AZ2	230	151	66	26	5	13-38	19.2
AZ3	49	24	49	25	6.7	11-37	26.8
NZ1	172	111	64	26	4.4	10-39	16.9
JX2	246	191	78	29	4.1	13-39	14.1
RX1	104	78	75	18	1.6	14-22	8.8
AX1	163	86	53	27	4.3	14-40	15.9
GZ1	108	83	77	26	5.4	13-40	20.8
Px1	247	154	62	26	4.6	12-40	17.7
AZ1	52	38	73	30	4.4	22-43	14.7
UX1	27	23	85	25	4	15-30	16
GZ2	47	38	81	26	4.5	15-33	17.3
<u>Blackline</u>							
WIP3	158	66	42	26	5	12-35	19.2
WIP2	10	6	60	25	2.3	23-99	9.2
<u>Control</u>							
UX022	71	59	83	23	3.7	14-29	16.1
<u>English</u>							
Vina	14	10	71	18	3.7	13-24	20.5
Sunland	64	20	31	26	3.8	18-31	14.6
Totals	1868	1225	66	25			

TABLE 2

Walnut Rootstock Orchard Trials-Yuba County. Established May 2005. Randomized complete block design with 4 clonal and 1 seedling ‘Paradox’ rootstock. Twelve single tree replicates. All rootstocks generally grew well except AZ2. Midday stem water potential measurements suggest that replants and adjacent orchard trees were generally in a similar range to the replants. AZ2 did not grow well despite having midday stem water potentials in a similar range to adjacent trees and the other replants. There was 100% survival of all replant sources.					
Treatment	June 2005 Growth Rating*	June 2005 Diameter (mm)	December 2005 Diameter (mm)	November 2006 Diameter (mm)	% Change (June 2005 to November 2006)
VX211	3.0	29.0	31.2	42.5 a	+46.6
AZ2	2.0	26.9	28.5	26.2 c	-2.6
NZ1	2.9	23.5	25.8	35.6 b	+51.4
JX2	2.2	23.7	26.4	35.2 b	+48.5
CONTROL	1.4	20.5	22.4	30.0 bc	+46.3

*rootstocks were headed near or below 2005 diameter measurement point. 2006 measurement is on new shoot growing above previous cut.
*Growth rating descriptions:
0 No growth
1 Just breaking
2 Moderate growth
3 Vigorous growth

TABLE 3

Field performance of clonal ‘Paradox’ walnut hybrids, Northern California black walnut and Chinese wingnut rootstocks in non-infested soil and soil infested with <i>Phytophthora</i> .				
Clone (or species)	Maternal background of hybrid (or species of standard)	Soil treatment (January 2006)	Incidence of crown rot (%)	
10 AX1	<i>californica</i>	Control	0 c	
		<i>P. citricola</i>	4 c	
AZ2	(major x <i>hindsii</i>)x <i>nigra</i>	Control	0 c	
		<i>P. citricola</i>	0 c	
NZ1	(major x <i>hindsii</i>)x <i>nigra</i>	Control	0 c	
		<i>P. citricola</i>	0 c	
15 GZ1	<i>hindsii</i>	Control	0 c	
		<i>P. citricola</i>	4 c	
JX2	<i>hindsii</i>	Control	0 c	
		<i>P. citricola</i>	0 c	
PX1	<i>hindsii</i>	Control	0 c	
		<i>P. citricola</i>	8 bc	
20 VX211	<i>hindsii</i>	Control	0 c	
		<i>P. citricola</i>	0 c	
RX1	<i>microcarpa</i>	Control	0 c	
		<i>P. citricola</i>	0 c	
WIP3	<i>hindsii</i> x <i>regia</i>	Control	0 c	
		<i>P. citricola</i>	8 bc	
25 (NCB)	(<i>J. hindsii</i>)	Control	16 b	
		<i>P. citricola</i>	62 a	
(Wing-nut)	(<i>Pt. stenopiera</i>)	Control	0 c	
		<i>P. citricola</i>	0 c	
30	Clone (or species)	Percent of trunk circ. Necrotic	Incidence of tree mortality (%)	Increase in trunk circ. (mm)
	AX1	0 c	0 c	163 c
		1 c	0 c	146 cde
	AZ2	0 c	0 c	116 fg
		0 c	0 c	117 fg
	NZ1	0 c	0 c	116 fg
		0 c	0 c	130 def
	GZ1	0 c	0 c	157 cd
		1 c	0 c	150 cd
	JX2	0 c	0 c	166 bc
		0 c	0 c	135 def
	PX1	0 c	0 c	169 bc
		1 c	0 c	157 cd
	VX211	0 c	0 c	191 b
		0 c	0 c	147 cde
	RX1	0 c	0 c	112 fg
		0 c	0 c	116 fg
	WIP3	0 c	0 c	100 g
		2 c	0 c	121 efg
	(NCB)	17 b	17 b	65 h
		59 a	59 a	57 h
	(Wingnut)	0 b	0 c	226 a
		0 b	0 c	193 b

¹All trees were planted May 2005. The assessments of crown rot and mortality were made 21 Nov. 2006. Means within a column and without letters in common are significantly different (Waller k ratio).

What we claim is:

1. A new and distinct variety of walnut rootstock plant designated ‘VX211’ as shown and described herein.

FIG. 1

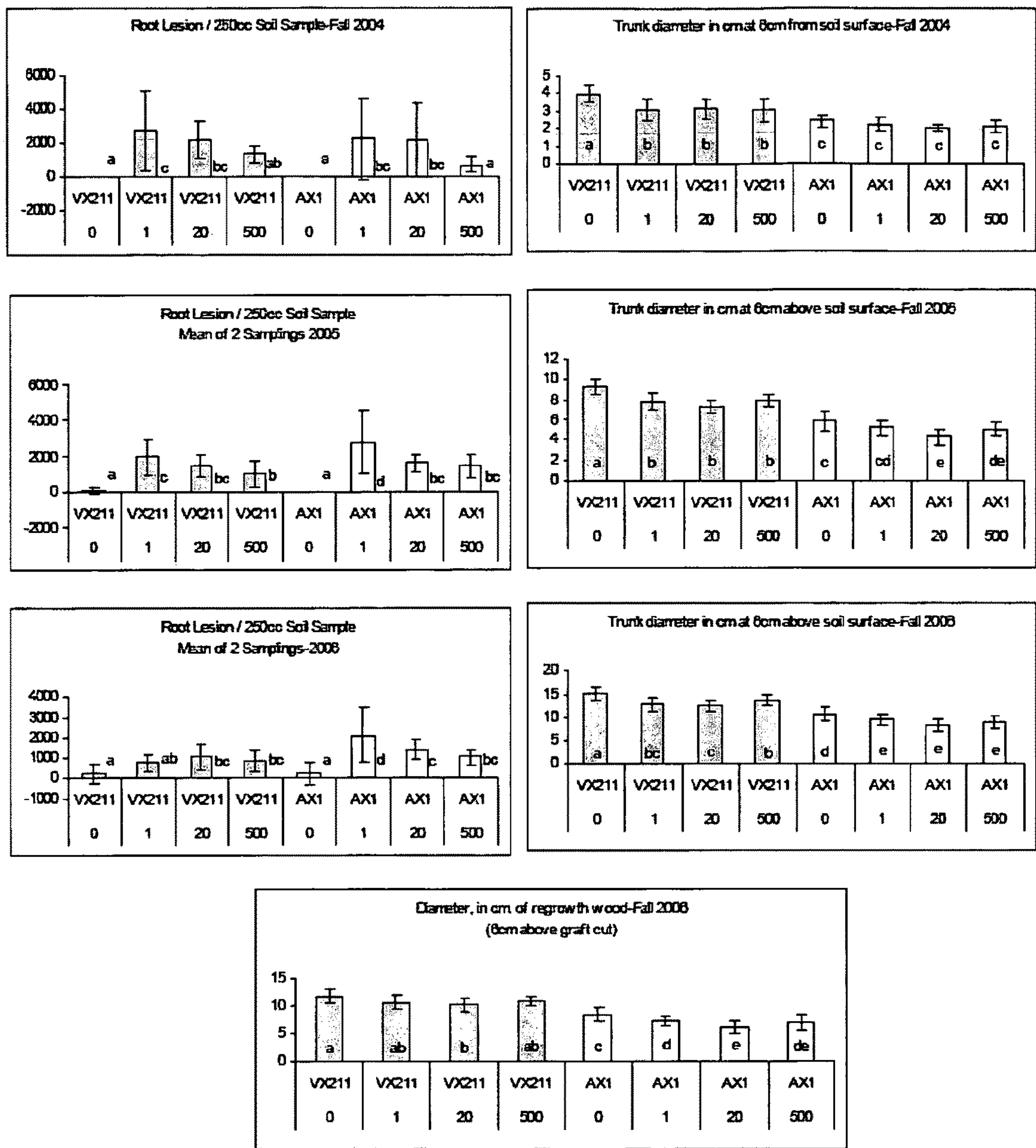


FIG. 2

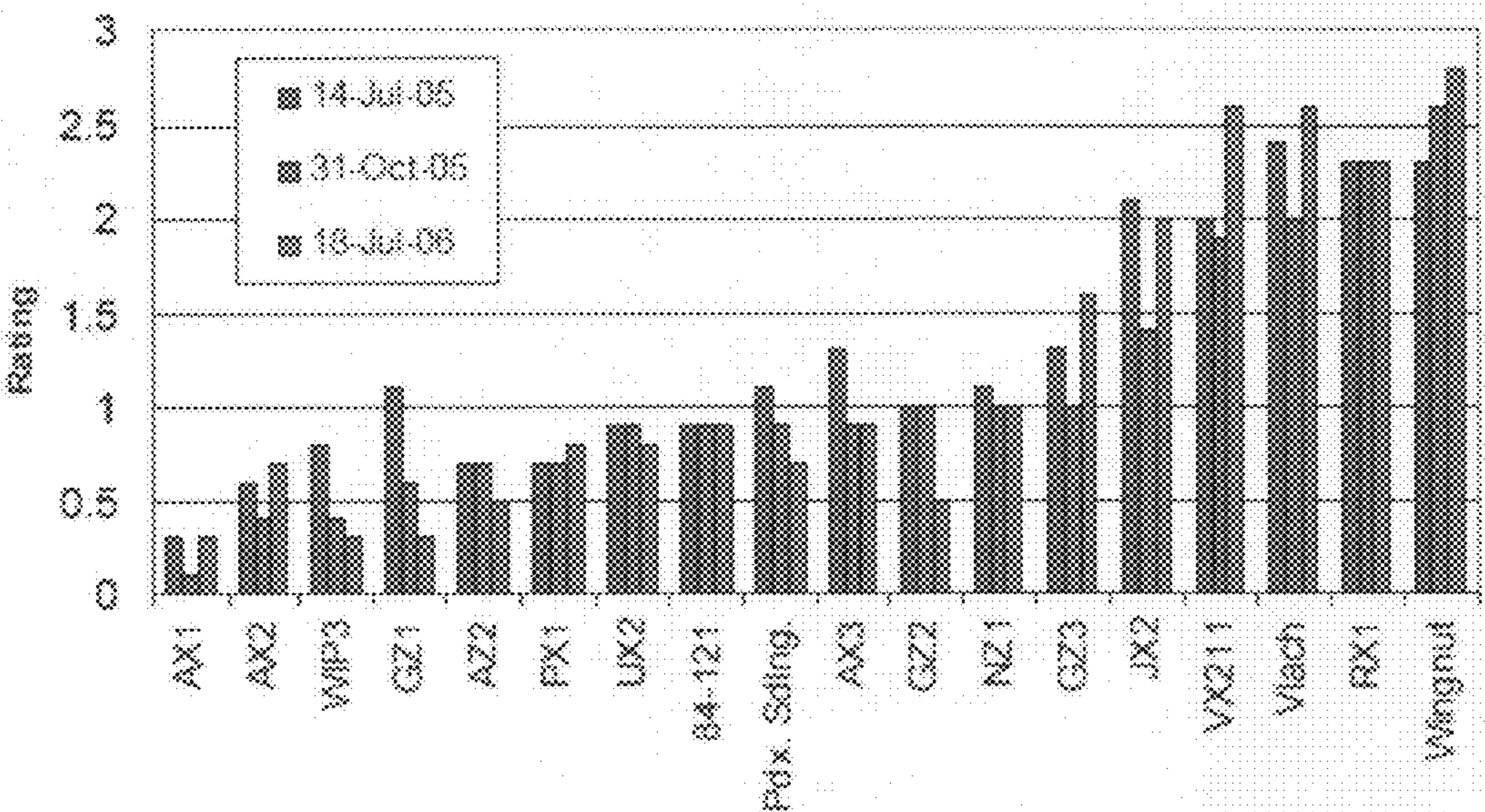


FIG. 3

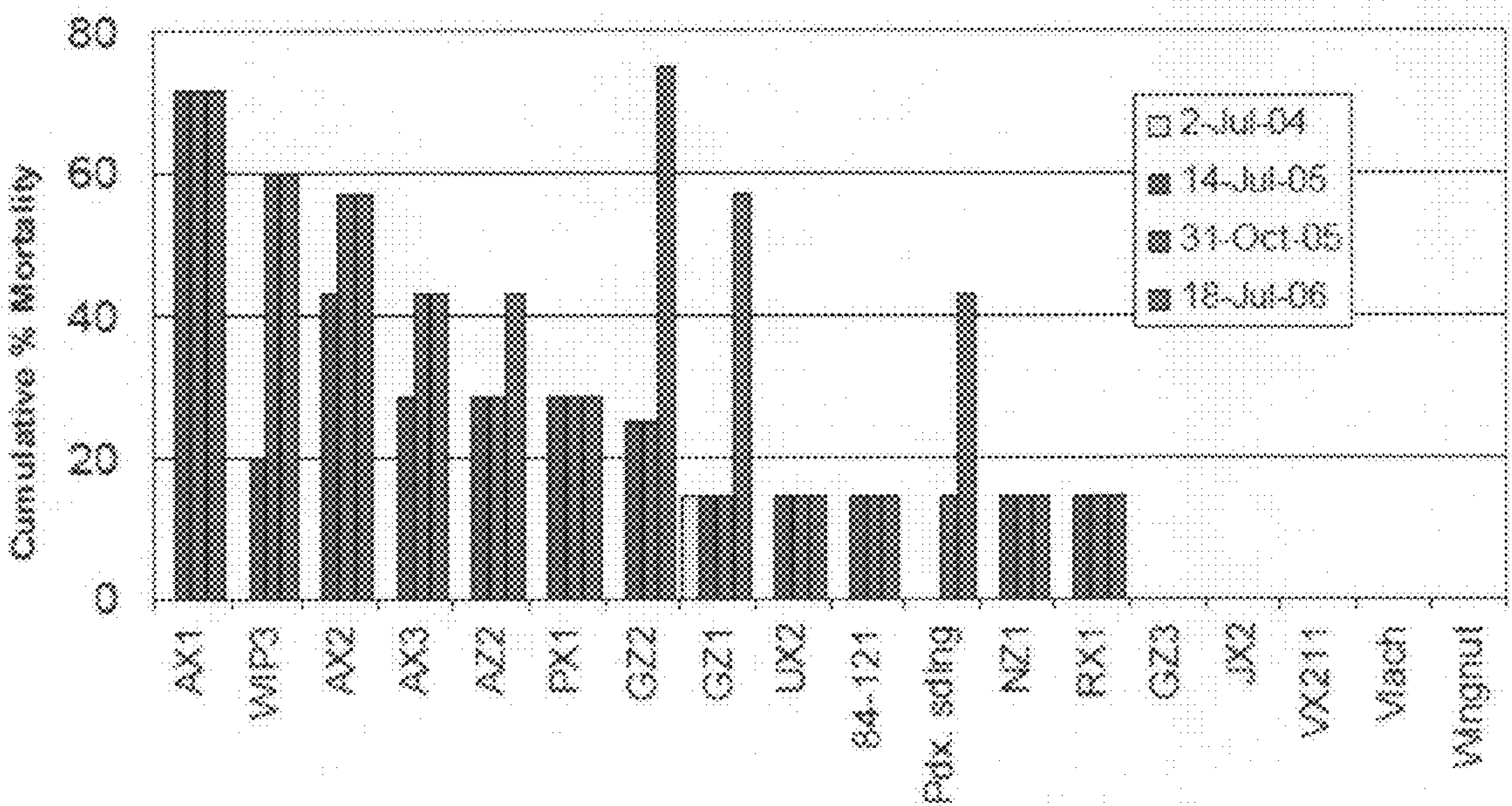


FIG. 4

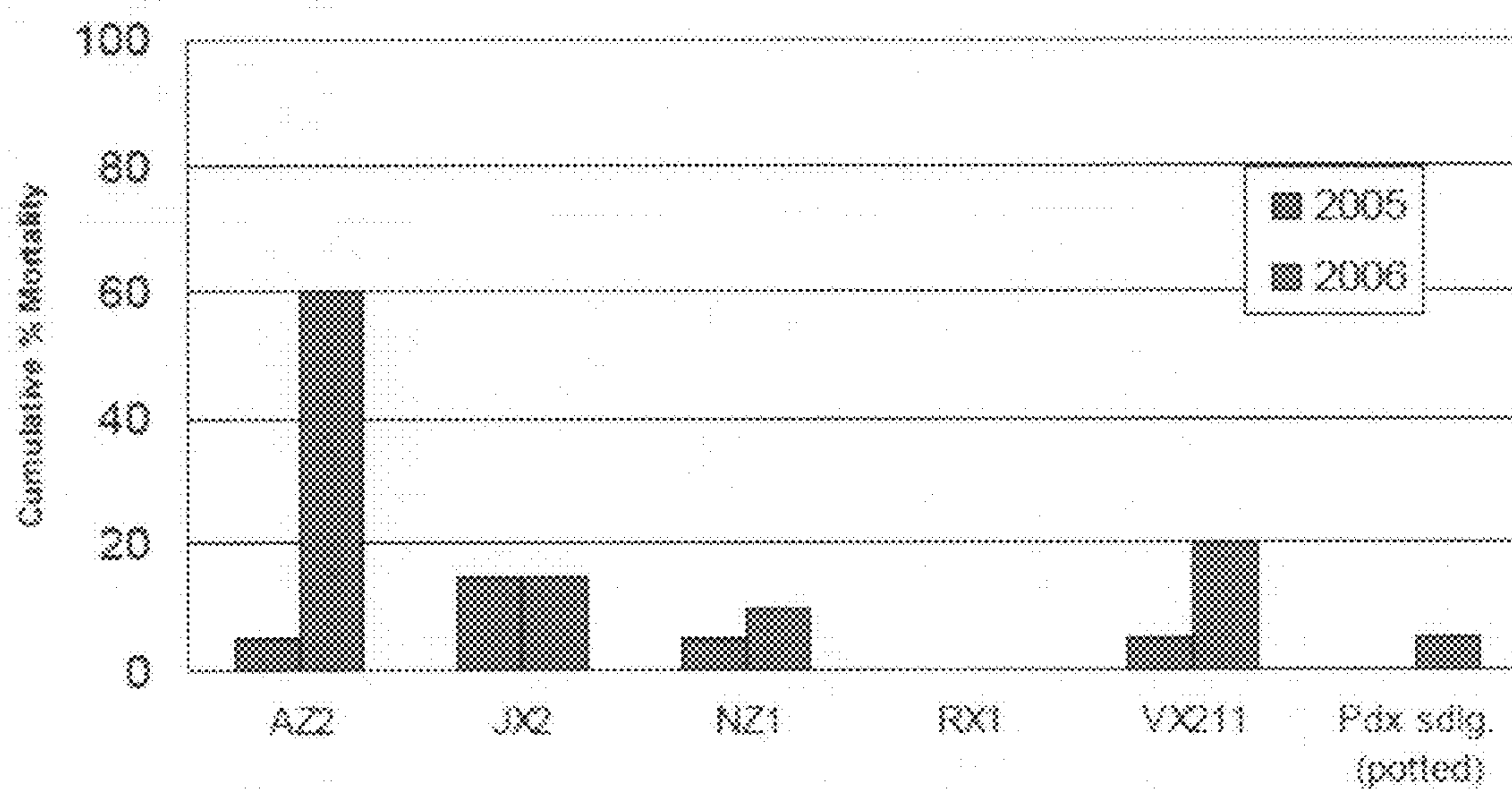


FIG. 5

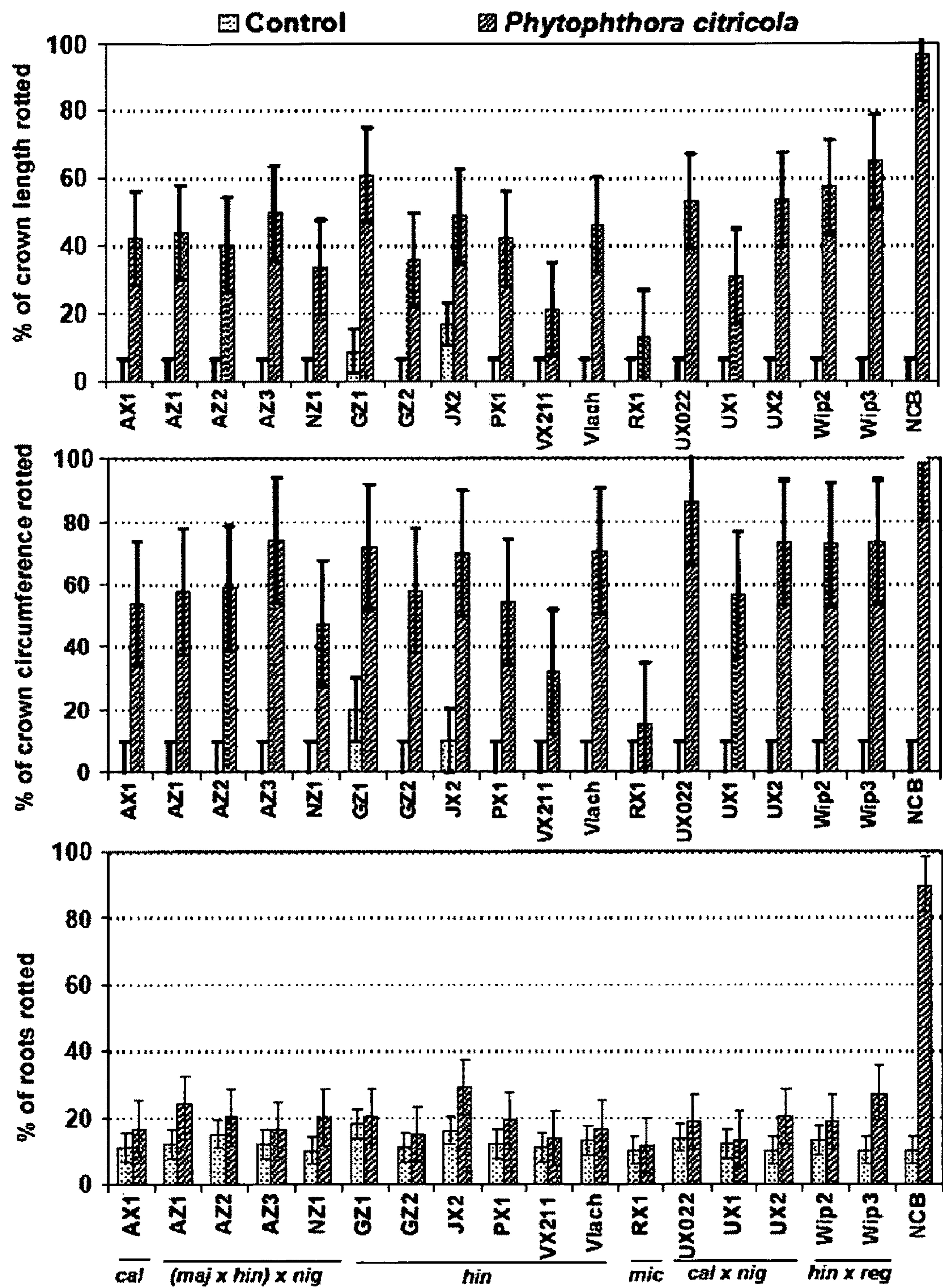


FIG. 6



FIG. 7



FIG. 8

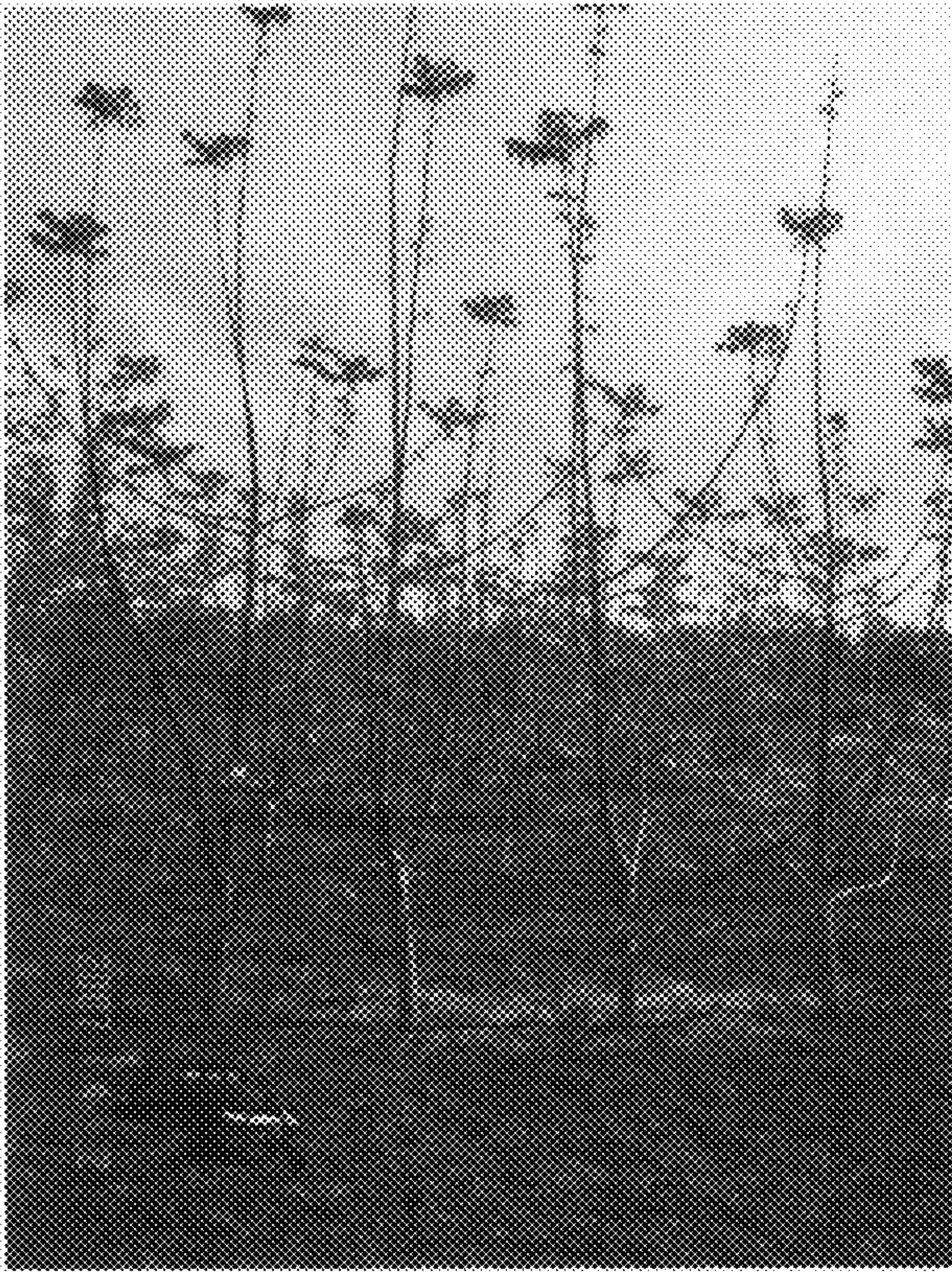


FIG. 9

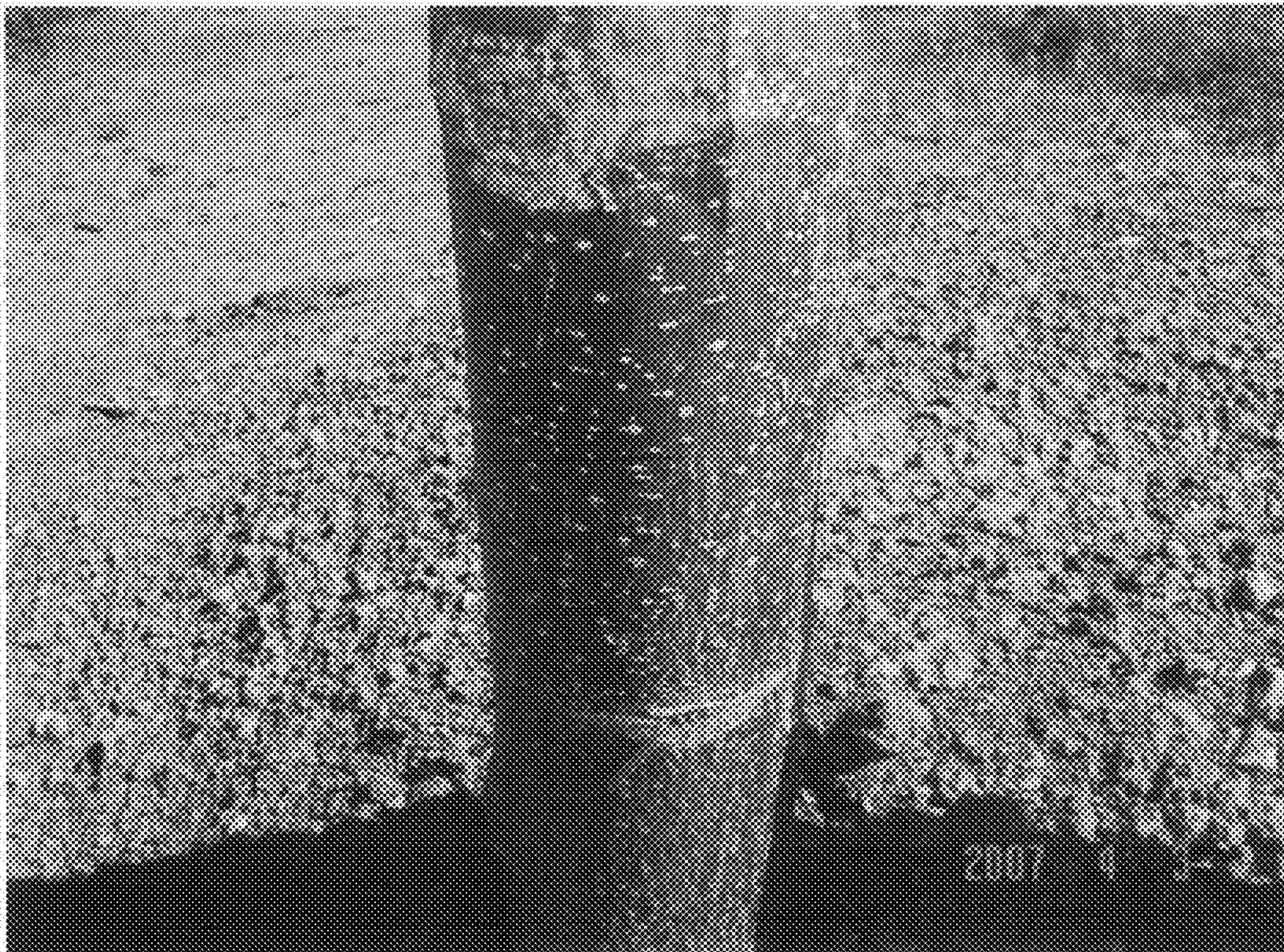


FIG. 10



FIG. 11



FIG. 12



FIG. 13



FIG. 14

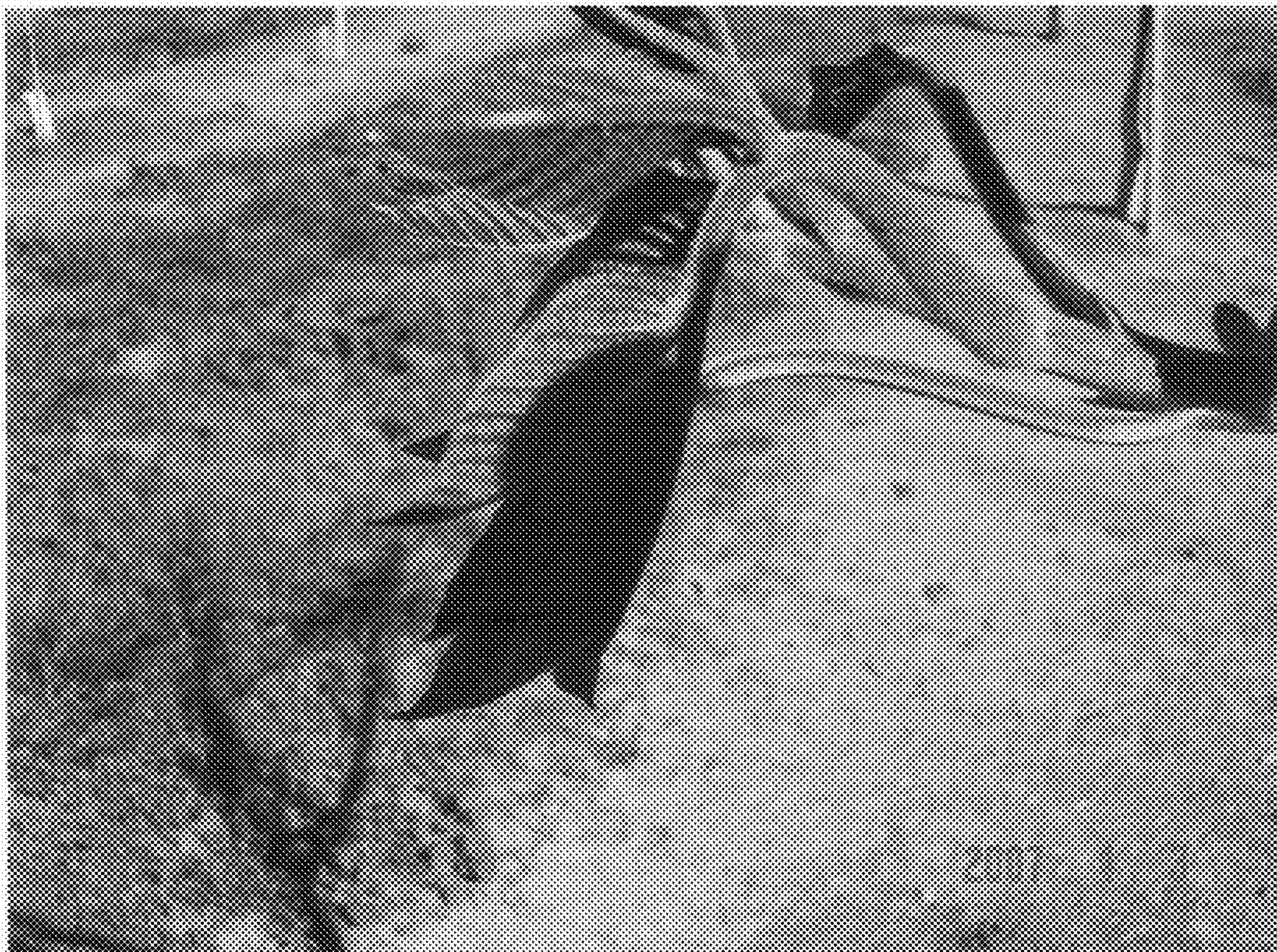


FIG. 15

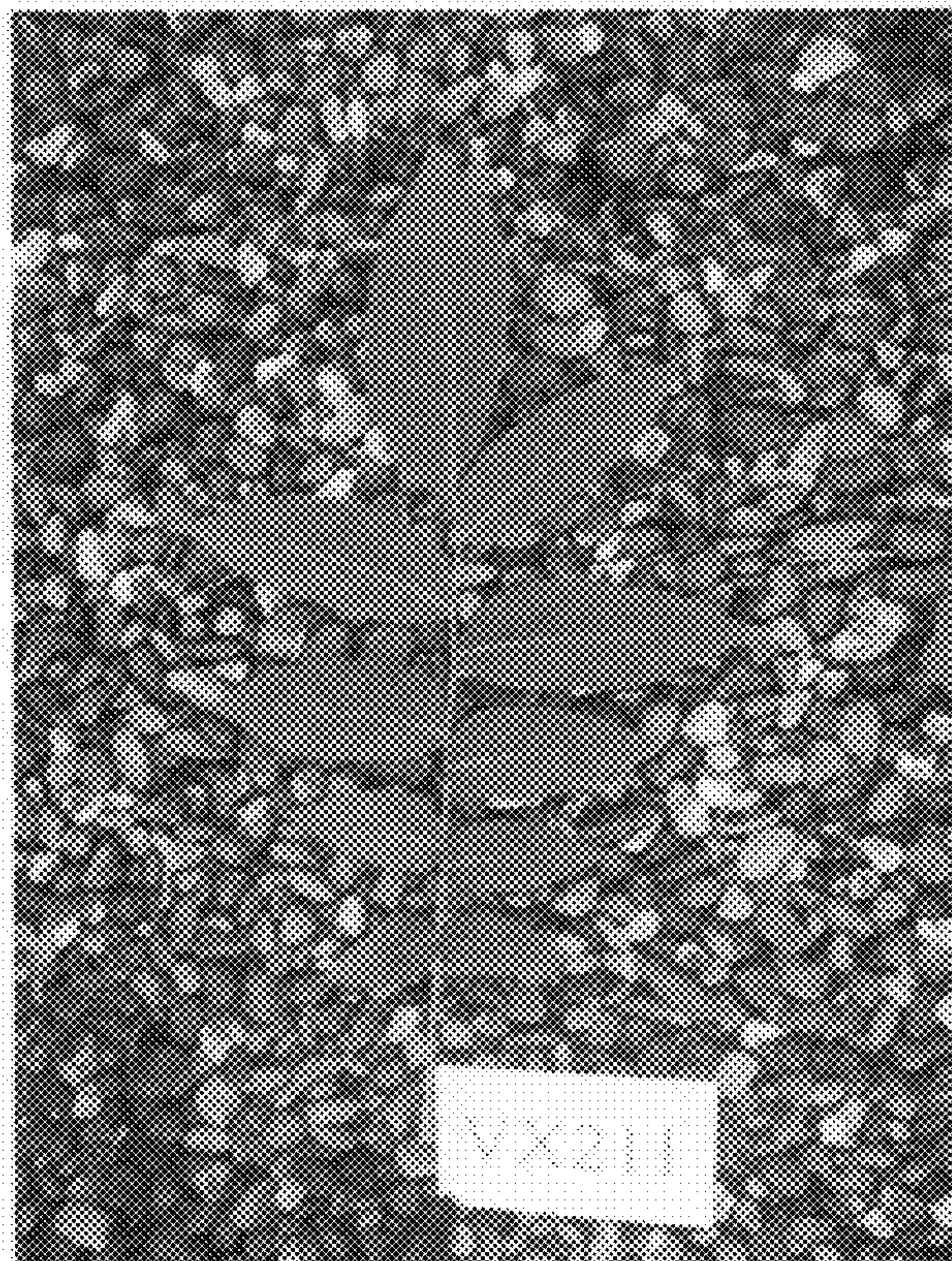


FIG. 16

